

Obscured, But Not Blind: Systematizing Observability in Confidential AI Training Clusters

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Abstract—Confidential computing is increasingly used in AI training clusters to protect sensitive data and model intellectual property from infrastructure operators and co-tenants. The same hardware isolation and end-to-end encryption that provide these guarantees also blind traditional intrusion detection systems (IDS): payloads are opaque, host-based monitoring cannot cross the Trusted Execution Environment (TEE) boundary, and RDMA kernel bypass removes most OS-level visibility. This is often read to mean that confidential clusters are unmonitorable. We argue that the real question is narrower and unanswered: what *can* an IDS see? We systematize the observability that survives encryption and the TEE boundary across three metadata layers: network, infrastructure, and workload. For each layer we establish, from first principles and prior work, the footprint an attack is predicted to leave, and validate key components of this predictive surface with worked-example measurements from two testbeds: a GPU training cluster and a DPU testbed. We show that resource attacks such as RDMA flooding and bandwidth squatting are predicted to remain visible from the architecturally isolated DPU vantage—and validate this under IPsec—while training-disruption attacks predictably surface as outliers in collective timing. We also define the theoretical limits of metadata-only detection, from the noise floor of covert channels to attacks that leave no footprint by design, and we close with a research agenda and design guidance for cross-layer, DPU-vantage IDS in confidential AI clusters.

1. INTRODUCTION

The transition of AI training clusters to confidential computing architectures, such as NVIDIA H100 Confidential Computing, Intel TDX, and AMD SEV-SNP, represents a critical security advancement [1]. By isolating workloads in Trusted Execution Environments (TEEs) and enforcing end-to-end encryption, these architectures protect sensitive training data and model intellectual property from compromised infrastructure and malicious co-tenants. However, this same isolation creates an observability paradox: the encryption and hardware boundaries that protect the workload simultaneously blind traditional intrusion detection systems (IDS) [2]. Payload inspection becomes impossible, host-based monitoring cannot cross TEE boundaries, and RDMA kernel bypass eliminates most OS-level network visibility.

This is often taken to mean that confidential clusters are effectively unmonitorable. We argue instead that the field lacks a systematic account of what observability actually remains. The relevant question is not whether the payload can be read, which it cannot, but rather: for each class of attack against an AI cluster, what metadata footprint survives encryption and the TEE boundary, from which vantage point is it observable, and where does that footprint vanish entirely?

This paper systematizes the answer. Rather than presenting a single deployed detector or producing ROC curves, we treat observability itself as the object of study. We proceed in two steps for each metadata layer. First, drawing on prior work in RDMA security [3], [4], congestion control [5], [6], distributed-training attacks [7]–[11], and cluster observability [12], [13], together with first-principles reasoning about the protocols and hardware, we establish the predictive footprint each attack is expected to leave. Second, we present worked-example measurements from two complementary testbeds that serve to validate these key predictions.

Building upon the preliminary conceptual framework from our prior conference paper [14], our primary contributions are:

1. **A predictive observability surface.** Across the network, infrastructure, and workload layers, we establish from literature and first principles the metadata footprint each attack is expected to leave, systematizing the theoretical detection capabilities.
2. **Empirical validation.** For each of the three metadata layers, we provide illustrative measurements from two vantages (an isolated DPU and an in-TEE workload exporter) that validate key components of these predictions in practice.
3. **The boundary of metadata-only detection.** We quantify the practical noise floor for covert timing channels and delineate the attacks that remain invisible by construction to a metadata-only IDS, without payload access.
4. **A research agenda and defender guidance.** We convert the gaps in this surface into a concrete open-problems agenda and design guidance for cross-layer, DPU-vantage IDS.

The remainder of this paper is organized as follows. Section 2 establishes the background, adversary model, and attack

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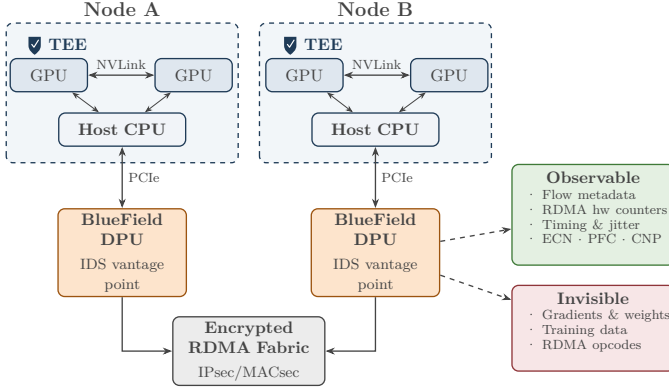


Fig. 1. Confidential AI training cluster architecture. Each node’s host CPU (confidential VM) and GPUs execute within a TEE boundary (dashed). The BlueField DPU sits outside the TEE on the network path, serving as the IDS vantage point: it observes flow metadata and hardware counters while remaining blind to encrypted payloads and model data.

taxonomy. Section 3 describes our methodology and testbed setup. Section 4 summarizes the observability surface and the evidence. Section 5 defines the three metadata layers (network, infrastructure, and workload). Section 6 presents the attack observability profiles, detailing the cross-layer footprint of each attack. Section 7 establishes the limits of metadata-only detection. Section 8 derives the research agenda and defender guidance. Section 9 reviews related work, and Section 10 concludes.

2. BACKGROUND AND THREAT MODEL

This work systematizes observability in confidential AI training clusters and is grounded in the conceptual framework of our prior conference paper [14], whose adversary model, attack taxonomy, and detectability tiers we adopt and extend with literature anchoring and empirical evidence.

2.1 Adversary Model

We consider three tiers of adversaries operating within a confidential AI training cluster:

1. **Compromised worker node:** An insider or malicious co-tenant who controls one or more GPU nodes and can deviate arbitrarily from the distributed training protocol.
2. **Co-located adversary:** An attacker sharing physical infrastructure who can inject or observe traffic on shared network links.
3. **Privileged infrastructure component:** A rogue administrator or compromised hypervisor. Crucially, this adversary controls the host OS but cannot breach the hardware-enforced TEE boundary protecting the AI workload.

The defining constraint of our environment is that confidential computing and end-to-end encryption prevent the adversary (and the IDS) from reading or modifying

encrypted payloads. The adversary can, however, observe and manipulate metadata: timing, flow sizes, RDMA congestion signals, and routing (Figure 1). This is precisely the surface our systematization characterizes.

2.2 Attack Taxonomy

We adopt the same 13 attacks, grouped by the subsystem they target. A1 (RDMA flooding / resource exhaustion) and A2 (bandwidth squatting / congestion injection) exploit the RDMA fabric, while A3 (BGP / routing manipulation) alters fabric routing to create path asymmetry or latency. A4 (Byzantine node slowdown), A5 (gradient poisoning), A6 (training data poisoning), and A7 (AllReduce / collective interference) target the distributed-training protocol. A8 (TEE side-channel attacks), A9 (memory bus snooping), and A10 (passive timing observation) attempt to leak data, while A11 (rogue GPU node behavior), A12 (covert timing channels), and A13 (model / IP theft via side channels) exploit allocated resources.

Each attack is anchored to the literature that motivates its expected footprint. The plausibility and stealth of the fabric attacks (A1, A2) are established by ReDMark [3] and the low-rate LoRDMA attack [4], with their expected congestion footprint following from DCQCN [5] and RoCEv2 deployment experience [6]. The training-protocol poisoning variants (A4–A7) are motivated by the Byzantine-ML literature—Krum [7], the hidden vulnerability of distributed learning [8], “A Little Is Enough” [9], local model poisoning [10], and backdoor injection [11]. The detectability of the data-leak attacks (A8–A10) via hardware performance counters is informed by prior counter-based malware detection [15], and the side-channel variants (A11–A13) by NVLink contention channels [16], [17]. Section 4 recaps the full attack set with its predicted detectability tier and marks which attacks our worked examples actually demonstrate (Table 1).

2.3 Detectability Tiers

These attacks partition into three predicted detectability tiers: **Tier 1**, strongly detectable via metadata and timing; **Tier 2**, partially detectable only with specialized monitoring; and **Tier 3**, fundamentally undetectable by a metadata-only IDS. These tiers are predictions. A central purpose of this systematization is to confront them with first-principles analysis and to selectively validate them using empirical evidence from our testbeds.

3. METHODOLOGY AND EVIDENCE BASE

This section describes the evidence base behind our worked examples used to validate the predictive observability surface.

3.1 The Evidence Base: Two Vantages

Our worked examples are drawn from two complementary testbeds (Figure 1), each instantiating a distinct vantage and a distinct threat perspective from the adversary model

of Section 2.1. The GPU training cluster is the tenant’s view: telemetry is exported from inside the TEE and the DPU management plane is out of reach, which is exactly the position from which a compromised worker node is observed. The BlueField-3 testbed is the infrastructure operator’s view: full access to the DPU’s isolated management plane but no insight into workload internals, which is the position from which a co-located traffic injector and a privileged-but-TEE-blocked host are observed. This separation allows us to study each adversary from the vantage that can actually see it; mapping the metadata layers to those vantages is precisely the systematization’s task.

The workload vantage (GPU training cluster). An NVIDIA H100 80GB cluster, managed by the Slurm workload manager, running PyTorch 2.5.1 with CUDA 12.2. Distributed experiments used two-node allocations (`--nodes=2 --gres=gpu:1`) running a Distributed Data Parallel (DDP) ResNet-18 workload. This vantage provides high-fidelity workload telemetry (GPU utilization via DCGM, NCCL collective timing) but reflects a tenant threat model: the DPU management plane was inaccessible from the compute nodes. It is the natural vantage for the workload layer (W).

The network and DPU vantage (BlueField-3 testbed). Two back-to-back BlueField-3 (BF3) DPUs, with two independent x86 hosts directly connected DPU-to-DPU via 200 GbE ConnectX-7 links and no intermediate switch. This vantage provides full SSH access to the DPU ARM cores (eswitch representor statistics, OVS flow tables, `devlink` metrics, DOCA telemetry, and hardware crypto/IPsec counters), and is the natural vantage for the network (N) and infrastructure (H) layers. ECN is enabled and PFC disabled on this fabric; 802.3x global pause is observable.

This mapping, in which the DPU vantage observes N and H while the in-TEE exporter observes W, recurs throughout the paper and motivates the cross-vantage fusion argument of Section 8. Crucially, because the BF3 testbed is back-to-back, it lacks an intermediate switch queue. Consequently, congestion-control signals that rely on switch-level queueing thresholds (like ECN and RoCEv2 CNPs) will naturally not be triggered on this specific topology, even under heavy load, though they are predicted to be observable on a standard switched fabric.

3.2 Measurement Protocol

Each attack-severity combination was repeated at least five times (ten for baselines). A run was included in the analysis only if its telemetry window overlapped the injected attack window. All telemetry used UTC timestamps at millisecond precision, with hardware counters polled at roughly 10 Hz.

4. OBSERVABILITY SURFACE SUMMARY

The surviving metadata partitions into three layers: network-layer signals (N), infrastructure-layer signals (H), and workload-layer signals (W). The two vantages of

Section 3.1 observe these layers asymmetrically: the isolated DPU sees the network and infrastructure layers, while the trusted in-TEE exporter sees the workload layer. Section 5 defines the signals in each layer, which form the vocabulary for the attack profiles in Section 6.

Table 1 recaps the full attack set with its predicted detectability tier, forming the complete predictive map that the rest of this section and Section 6 systematically step through and selectively validate.

Rather than enumerate every attack against every signal, Section 6 focuses on the core attacks for which our testbeds provide strong empirical validation of the predictive surface. Seven attacks (A1, A2, A4, A5, A7, A11, A12) carry such measured evidence; the remainder (A3, A6, A8–A10, A13) serve as theoretical limits or open problems, treated in Sections 7 and 8.

Structurally, the validated signals cluster predictably by layer and vantage: network and infrastructure footprints concentrate in the resource attacks (A1, A2) seen from the DPU, while workload footprints concentrate in the training-disruption attacks (A4, A7) seen from the in-TEE exporter. The sections that follow define the metadata layers and then walk the observability surface one attack at a time.

5. THE METADATA LAYERS

The observability surface is built upon three distinct metadata layers. Each layer is observed from a specific vantage point and captures a different aspect of the cluster’s behavior.

5.1 Network-Layer Signals (N)

The network layer is where confidential computing’s blinding is most often assumed to be total, yet first principles say otherwise. Transport encryption (IPsec ESP) conceals the payload and the inner transport header, but it cannot conceal the existence, volume, or timing of packets on the wire. Flow size, packet counts, duration, and burstiness are properties of the traffic envelope, not its contents, and therefore survive. For RDMA fabrics specifically, the DCQCN control loop [5] predicts a characteristic congestion cascade under load: switch queues cross an ECN threshold, receivers return RoCEv2 CNPs, and sustained pressure escalates to PFC pause frames. Table 2 catalogs the signals of all three metadata layers; its network-layer rows (N) summarize what is observable on the wire and at the DPU.

5.2 Infrastructure-Layer Signals (H)

The infrastructure layer is the strategic core of the thesis. The DPU sits on the data path but outside the host TEE, giving it an architecturally isolated vantage: a compromised host OS cannot suppress or forge DPU-side telemetry. First principles predict that the traffic envelope re-materializes here as hardware counters. Most importantly, because these counters measure the physical wire, they should be encryption-invariant: enabling IPsec changes the payload

TABLE 1
PREDICTED DETECTABILITY TIER FOR THE 13 ATTACKS.

ID	Attack	Target / mechanism	Pred. tier
A1	RDMA flooding / resource exhaustion	RDMA fabric	1
A2	Bandwidth squatting / congestion injection	RDMA fabric	1
A3	BGP / routing manipulation	Fabric routing	1
A4	Byzantine node slowdown	Training protocol	1
A5	Gradient poisoning (norm-inflated)	Training protocol	2
A6	Training data poisoning	Training protocol	3
A7	AllReduce / collective interference	Training protocol	1
A8	TEE side-channel attacks	Confidentiality (TEE)	2
A9	Memory bus snooping	Confidentiality (physical)	3
A10	Passive timing observation	Confidentiality (passive)	3
A11	Rogue GPU node behavior	Allocated resources	1
A12	Covert timing channels	Allocated resources	3
A13	Model / IP theft via side channels	Allocated resources	3

but not the byte and packet totals, save for a small, predictable ESP framing overhead. The infrastructure-layer rows (H) of Table 2 list these signals, delineating those accessible from the isolated DPU vantage and those residing inside the host TEE boundary.

5.3 Workload-Layer Signals (W)

The workload layer is observed not on the wire but through a trusted agent inside the TEE (e.g., DCGM or NCCL profiling) that exports macroscopic behavior without revealing payload semantics. Data-parallel training is highly periodic and symmetric: every rank executes the same step and meets at a global synchronization barrier, so GPU utilization is near-uniform across ranks and collective timing is tightly clustered. Any deviation from this symmetry should stand out as a statistical outlier against an exceptionally quiet baseline. The workload-layer rows (W) of Table 2 detail the signals exported by a trusted agent.

6. ATTACK OBSERVABILITY PROFILES

This section details the cross-layer footprint of each attack. For each attack, we describe the expected mechanism and present the empirical evidence gathered from our testbeds.

6.1 RDMA Flooding (A1)

RDMA flooding targets the fabric by exhausting link capacity or switch buffers. Because this attack operates purely at the transport layer, its expected footprint is concentrated in network and infrastructure volume signals: we expect a massive increase in flow size (N2) and link utilization (H2). If the flooding successfully induces fabric congestion, it should also trigger the standard DCQCN congestion cascade, visible as ECN marks (N6), RoCEv2 CNPs (N8), and eventually PFC or link-level pause frames (N9). Workload-layer signals (W1–W9) are not expected to move unless the flooding severely starves concurrent training traffic.

In our experiment on the BF3 vantage, five accepted low-severity `ib_write_bw` repetitions moved approximately

448.9 GB globally during the 60 s attack window, providing a strong validation of the predicted flow volume (N2) and physical link utilization (H2) signals.

The attack also produced a positive 802.3x global-pause signal (N9) on both host and DPU vantages, demonstrating genuine link-level backpressure. While ECN (N6) and RoCEv2 CNP (N8) are predicted to signal congestion on a standard switched fabric, they naturally remain inactive on our back-to-back testbed since there is no switch queue to cross an ECN threshold. This validates the volume-based predictions for A1 in a low-severity, single-path regime.

6.2 Bandwidth Squatting (A2)

Bandwidth squatting injects unauthorized traffic to steal capacity or induce congestion without necessarily flooding the link entirely. Like RDMA flooding, it is expected to generate strong flow volume (N2) and link utilization (H2) signals. The stealthier nature of squatting makes buffer pressure indicators—such as NIC queue depth (H1) and packet drops (H3)—critical for detection, alongside DPU representor counters (H5) that reveal unauthorized flow state. Congestion markers (N6, N8, N9) should only appear if the squatting genuinely pushes switch queues past their thresholds.

In our experiments across fifteen single-flow repetitions plus one incast repetition (fixed-rate UDP cross-traffic) on the BF3 testbed, A2 validated the predictions for byte/packet movement (N2) and link utilization (H2) in every run (15/15); this traffic-volume signal was large at all three configured offered rates, plainly visible even at the lowest setting (Figure 2(a)). The attack also drove DPU representor counter movement (H5).

Buffer-pressure counters were positive in 12 of 15 single-flow runs and on both links of the incast upgrade: packet drops (H3) emerged as a strong signal, with NIC queue depth (H1, including `rx_out_of_buffer` and DPU `vport_out_of_buffer`) providing supporting evidence; this buffer-pressure signal is positive but noisier than the volume signal (Figure 2(b)). As with A1, the congestion

TABLE 2
METADATA-LAYER SIGNALS AND THEIR VANTAGES.

ID	Signal	Description
<i>Network layer (N) — observed on the wire and at the DPU</i>		
N1	Packet inter-arrival time	Spacing between successive packets; carries timing structure even when payloads are encrypted.
N2	Flow size	Total bytes and packets exchanged in a flow; the core volume signal.
N3	Flow duration	Wall-clock lifetime of a flow, from first to last packet.
N4	Packet size distribution	Distribution of on-wire packet lengths (modulo ESP padding).
N5	Burstiness	Temporal clustering of transmissions versus a smooth rate.
N6	ECN marking	Explicit Congestion Notification bits set by a switch when its queue crosses a threshold.
N8	RoCEv2 CNP	Congestion Notification Packets returned by a receiver under DCQCN.
N9	PFC / 802.3x pause	Priority and global link-level flow-control pause frames.
N10	RDMA retransmissions / errors	Retransmit and error events on the RDMA connection.
<i>Infrastructure layer (H) — on the DPU data path; H7–H9 inside the host TEE boundary</i>		
H1	NIC queue depth	Occupancy of NIC queues, including out-of-buffer events.
H2	Link utilization	Physical port byte and packet rate.
H3	Packet drops	Dropped-packet counters at the NIC and DPU ports.
H4	RDMA performance counters	RDMA operation counters (e.g., out-of-sequence, errors).
H5	DPU OVS / representor counters	eswitch representor statistics and OVS flow-table state on the DPU.
H6	DPU ARM CPU utilization	Compute load on the DPU control-plane cores.
H7	PCIe bandwidth	Host-to-device PCIe throughput.
H8	GPU memory bandwidth	HBM bandwidth utilization.
H9	NVLink bandwidth	Inter-GPU NVLink throughput.
H10	Host CPU / memory utilization	Host-level processor and memory usage.
H11	DPU crypto / IPsec counters	ESP and crypto-offload counters on the DPU.
<i>Workload layer (W) — exported by a trusted agent inside the TEE</i>		
W1	GPU utilization per node	Per-rank GPU busy fraction; near-uniform across ranks in healthy data-parallel training.
W2	AllReduce step timing	Wall-clock duration of each training step / collective.
W3	AllReduce bandwidth proxy	Effective collective bandwidth inferred from step timing and message size.
W4	Step duration jitter	Variance or coefficient of variation of step time.
W5	NCCL collective duration	Per-rank completion time of NCCL collectives; exposes rank asymmetry.
W6	Gradient norm	L2 norm of the local gradient, if explicitly exported by the framework.
W7	AllReduce rank-IP mapping	Mapping of collective ranks to network addresses.
W9	NCCL ring/tree topology	Structure of the collective communication graph.

family (N6, N8, N9, N10) behaves as expected for the back-to-back topology.

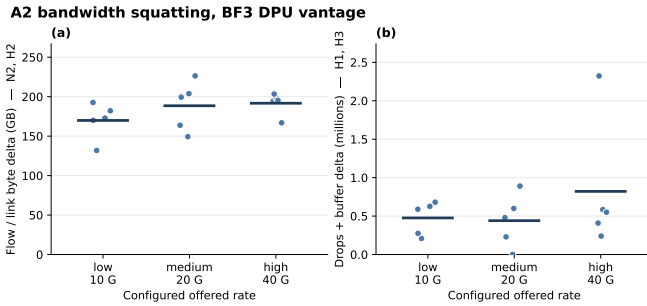


Fig. 2. A2 bandwidth squatting on the BF3 testbed (DPU vantage): attack-minus-baseline counter deltas across the configured offered rate ($n=15$, five per rate). (a) Traffic volume (flow/link bytes, N2/H2) and (b) buffer pressure (drops/buffer occupancy, H1/H3); points are per-run, bars are per-severity means. The volume signal is large and present in every run even at the lowest rate, while buffer pressure is positive but noisier (packet counts, also N2, track bytes and are omitted). The entire congestion-control cascade (ECN N6, RoCEv2 CNP N8, PFC/802.3x pause N9, retransmit/error N10) remains inactive due to the back-to-back topology, validating that the observed signals are not dependent on switch-level congestion; no achieved-rate monotonicity is claimed.

Encryption Invariance: The encryption-invariance ex-

pectation is the decisive worked example for infrastructure-layer observability. We re-ran A2 under host kernel XFRM transport-mode ESP (`cbc(aes) + hmac(sha256)`), fixing the offered rate at 800 Mbit/s UDP for both plaintext ($n=5$) and IPsec ($n=5$) runs so that throughput is not a confounder. Every IPsec run was verified to carry active ESP state with moving crypto counters (H11). All ten evaluated host- and DPU-side signal families were invariant. Host TX bytes (N2) measured 6.08e9 plaintext versus 6.32e9 under IPsec (ratio 1.04); DPU physical TX bytes (H2) 5.66e9 versus 5.87e9 (1.04); DPU representor vport RX bytes (H5) 5.64e9 versus 5.85e9 (1.04). The uniform ~3-4% increase precisely reflects ESP header, trailer, IV, and padding overhead, a benign metadata artifact rather than a loss of signal (Figure 3). The congestion family remained inactive in both conditions, confirming that its absence is topological rather than an artifact of encryption.

6.3 Byzantine Node Slowdown (A4)

A Byzantine slowdown attack maliciously delays a node’s participation in distributed training. Because data-parallel training relies on global synchronization barriers, this delay propagates to the entire workload. We expect this attack to be invisible at the network and infrastructure layers, as it merely spaces out packets without violating

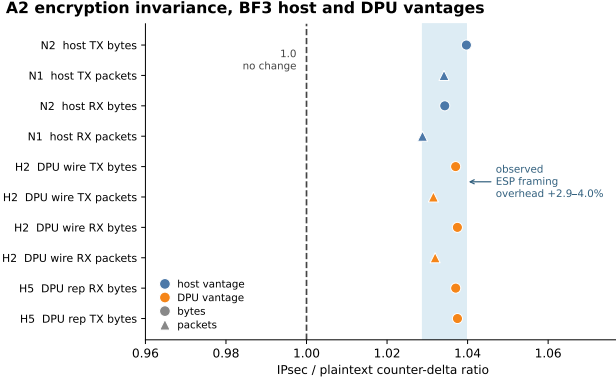


Fig. 3. Encryption invariance for A2 at a matched 800 Mbit/s offered rate ($n=5$ per condition). For each host- and DPU-side signal family we plot the IPsec/plaintext ratio of the attack-window counter delta; the dashed line marks perfect invariance (1.0). All ten families sit in a tight band (1.029–1.040, shaded), so the observable is preserved under host XFRM transport-mode ESP up to a *uniform* ~3–4% framing overhead (ESP header, trailer, IV, and padding)—no family collapses toward the null and none is inflated. Byte families carry slightly more overhead than packet families, as expected. The congestion-control family (ECN N6, CNP N8, pause N9) remains inactive in *both* conditions and is omitted here, confirming this behavior is topological and not an artifact of encryption.

transport protocols or exceeding link capacity. At the workload layer, however, the delay should severely break the synchronization symmetry: step timing (W2) will increase, per-rank collective duration (W5) will spread as fast ranks wait for the slow one, and GPU utilization (W1) will drop on the fast ranks as they idle.

In our experimental setup, using an in-process delay hook on a single rank of a 4-rank, 2-node job, mean step time rose monotonically from a 59.8 ms baseline to 516.6 ms at the highest severity, strongly validating the predicted W2 and W5 signals. The sharpest signature was per-rank collective timing (W5): mean AllReduce rank spread widened from 0.13 ms to 500.7 ms. GPU utilization (W1) fell monotonically as ranks idled at the barrier, and, counterintuitively, step-time jitter (W4) decreased (CV 0.61 to 0.047), because the constant injected delay became the dominant deterministic term (Figure 4).

6.4 Gradient Poisoning (A5)

Gradient poisoning attempts to corrupt the global model by submitting malicious updates. From a network and infrastructure perspective, the traffic envelope remains completely unchanged, as the size and timing of gradients are determined by the model architecture, not their values. Thus, N and H signals cannot detect it. At the workload layer, observability depends entirely on the nature of the poison. If the attacker scales the gradient vectors aggressively to overwhelm healthy updates (norm-inflated poisoning), the explicitly exported gradient L2 norm (W6) will reveal the outlier. If the attacker carefully preserves the norm (norm-correct direction poisoning), the attack leaves no metadata trace.

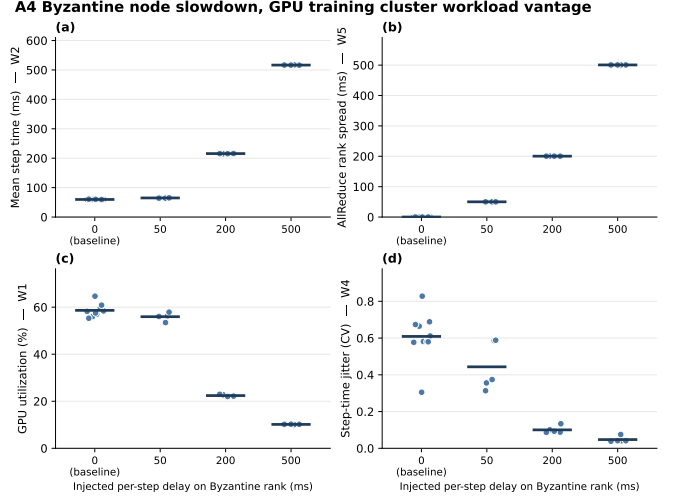


Fig. 4. A4 Byzantine node slowdown, GPU training cluster workload vantage, as a function of the injected per-step delay on Byzantine rank 2 ($n=10$ baseline, $n=5$ per severity). (a) mean step time (W2) rises monotonically; (b) AllReduce rank spread (W5)—the sharpest signature—widens from near zero to essentially the full injected delay as fast ranks wait for the slow one; (c) GPU utilization (W1) falls as honest ranks idle at the synchronization barrier; (d) step-time jitter (W4, coefficient of variation) *falls*, because the constant injected delay becomes the dominant deterministic term. Points are per-run; horizontal bars are per-severity means. The network and infrastructure vantages (N/H) remain unperturbed, as predicted: the attack only spaces packets without violating transport or exceeding link capacity.

In our experiment, via a DDP communication hook logging the local gradient L2 norm (W6), the poisoned rank’s reported norm tracked the injected inflation factor exactly (median $2\times/4\times/8\times$ its honest peers across the three severities), making it a clean outlier at every training step (Figure 5a). Crucially, the separation is peer-relative rather than a fixed threshold: both honest and poisoned norms decay as the model converges, so the late-training honest norm falls below the early-training poisoned norm (Figure 5b)—a static W6 alarm would either miss the attack or false-alarm on healthy early-training gradients. However, this visibility is entirely dependent on the trusted exporter. Furthermore, norm-correct direction poisoning, which alters the gradient direction while preserving the norm and all timing, remains fundamentally invisible to metadata-only detection, as predicted by our theoretical limits.

6.5 Collective Interference (A7)

Collective interference targets the AllReduce operations directly, disrupting the communication graph. If executed via network-level denial of service, this would surface similarly to A1 or A2. However, if executed logically within the framework—such as an attacker refusing to forward chunks in a ring AllReduce—the network fabric remains uncongested. The expected footprint is therefore isolated to the workload layer, appearing as rank-asymmetric collective timing (W5) and increased step duration (W2) without corresponding fabric drops or DPU pressure.

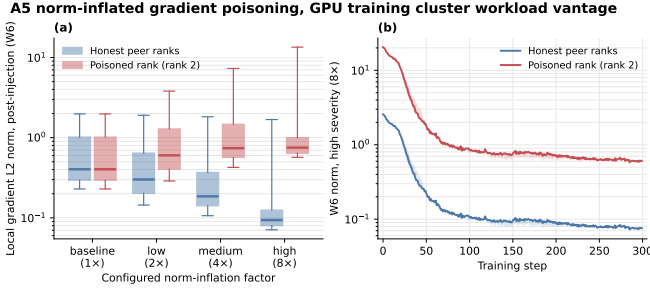


Fig. 5. A5 norm-inflated gradient poisoning, workload-exported W6 (post-injection local gradient L2 norm; $n=1500$ per group). (a) Per-step W6 distribution by configured inflation factor (box = median/IQR, whiskers 5th/95th percentile, log axis): at baseline the poisoned rank (rank 2) is indistinguishable from its peers, but it separates cleanly as the factor grows. (b) High-severity (8x) W6 trajectory: both honest and poisoned norms decay as the model converges, so the separation is peer-relative, not a fixed threshold — the late-training honest norm falls below the early-training poisoned norm. Applies only to norm-inflated poisoning; encrypted gradients and norm-correct direction poisoning preserve the norm and leave no W6 trace.

In our experiment, implemented with a non-privileged in-process fallback (since `tc netem` was denied by the tenant boundary), A7 produced rank-asymmetric collective timing (W5) and tail behavior (W2): p95 AllReduce rank spread rose from 0.33 ms baseline to 50.27 ms at high severity (Figure 6).

Crucially, this in-process delay generates strong workload timing signals yet zero fabric congestion or DPU effects. A workload-only IDS cannot distinguish this slow node from a network denial-of-service, which motivates cross-vantage fusion (Section 8).

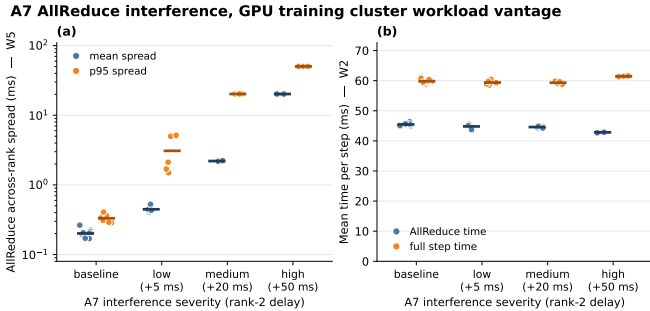


Fig. 6. A7 AllReduce interference, GPU training cluster workload vantage (per-run points, mean crossbar; $n=10$ baseline, $n=5$ each). (a) The signal: the across-rank AllReduce spread (W5) climbs monotonically with severity (p95 spread 0.33 $\rightarrow$ 50.27 ms), i.e. the collective becomes rank-asymmetric. (b) The discriminator: mean AllReduce time (≈ 45 ms) and mean full-step time (≈ 60 ms) stay flat — there is no monotonic mean shift (W2). The attack is in-process, so the fabric and DPU vantages stay null (zero congestion/drops/DPU pressure; reported in text, not plotted); only the rank-resolved spread reveals it.

6.6 Rogue GPU Node Behavior (A11)

A rogue node executes unauthorized workloads (e.g., cryptomining or secondary training) using allocated resources. This attack leaves a footprint across multiple layers,

requiring cross-vantage observation. At the network layer, unauthorized communication to external endpoints or non-training peers should create unexpected flows visible to the DPU (N2, H5). At the workload layer, the stolen compute cycles should break the symmetry of data-parallel training, reducing the rogue node’s effective training throughput and appearing as a drop in its GPU utilization (W1) dedicated to the legitimate job, accompanied by anomalous host CPU/memory usage (H10).

At the network and infrastructure layers, our rogue node footprint, emulated as an unexpected non-training UDP flow, surfaced at the DPU as representor byte movement (N2): the DPU host-facing representor carried 4.65 ± 0.08 MB during the 90 s attack window versus 0.14 MB at baseline ($\approx 32\times$, $n = 5$; Figure 7). The same flow churned the DPU OVS datapath (H5), which cycled through eight distinct megaflow states in every attack window against a single stable baseline state, and was supported by DPU ARM CPU utilization (H6). The offloaded megaflow did not preserve UDP-port detail, so the per-flow UDP-port hit count is a measured null. This network activity is invisible to the workload vantage.

A11 rogue node, BF3 DPU vantage

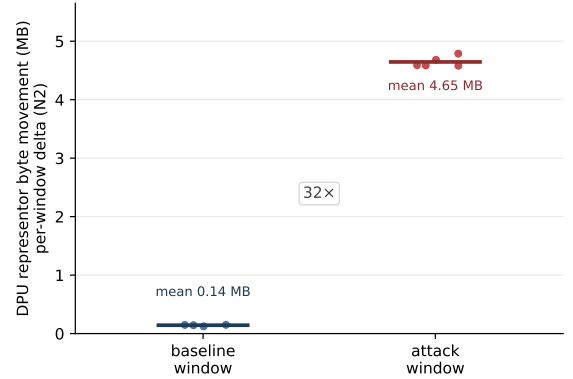


Fig. 7. A11 rogue node, BF3 DPU vantage. Per-run DPU host-facing representor byte movement (N2), baseline window vs. rogue-flow attack window (per-run points, mean crossbar; $n = 5$). The unexpected non-training flow lifts representor byte movement from a 0.14 MB baseline mean to a 4.65 MB attack mean ($\approx 32\times$). The same flow churns the DPU OVS datapath (H5; eight distinct megaflow states per attack window vs. one stable baseline state, reported in text). This network footprint is invisible to the workload vantage.

At the workload layer, emulating unauthorized background work on one rank of the data-parallel job left the rogue rank near its baseline GPU utilization (56.5% vs. a 58.5% all-rank baseline) while its honest peers stalled at the allreduce barrier and fell to $\approx 38\%$ (Figure 8a). This asymmetry raised the trusted exporter’s GPU-utilization peer spread (W1) monotonically with configured severity, from 1.97 pp at baseline ($n = 10$) to 18.70 pp at high severity ($n = 5$), with host CPU/memory peer spread (H10) separating from the noise only at high severity (Figure 8b). We claim no monotonicity in the *achieved* rogue rate, only that the workload-vantage spread tracks the configured severity.

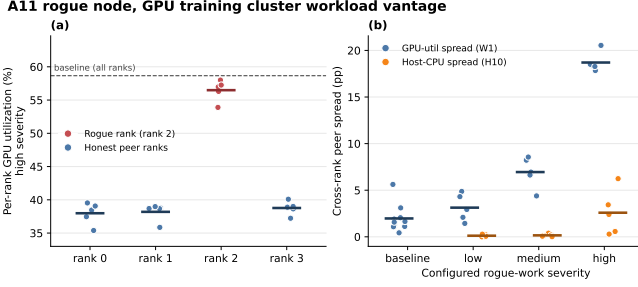


Fig. 8. A11 rogue node, GPU training cluster workload vantage (per-run points, mean crossbar). (a) Per-rank GPU utilization at high severity: the rogue rank holds near the all-rank baseline (dashed) while honest peers stall at the allreduce barrier and drop ($n = 5$ reps). (b) GPU-utilization peer spread (W1) rises with configured rogue-work severity, 1.97 pp baseline ($n = 10$) to 18.70 pp high ($n = 5$); host-CPU peer spread (H10) separates only at high severity. No achieved-rate monotonicity is claimed. This is a workload/host-exported view; the network footprint is the DPU-vantage figure.

7. THEORETICAL LIMITS AND FUTURE VALIDATION

Mapping where observability ends matters as much as mapping where it exists. We separate two kinds of limit: the quantitative noise floor that bounds a covert channel in practice, and the analytical limits where no metadata footprint is predicted to exist by design (or where empirical validation remains future work).

7.1 The Covert Channel Noise Floor (A12)

Covert timing channels (A12) represent a quantitative limit. On the BF3 vantage, a sender modulated a synthetic early/late burst signal in a 2 s slot. The cooperating receiver and host RDMA counters recovered the channel perfectly (BER 0, capacity 0.5 bit/s), but the DPU-vantage 0.5 s counter telemetry decoded near the noise floor: pooled DPU BER 0.41 (266/649), yielding a binary-symmetric-channel capacity bound of just 0.0118 bit/s (Figure 9). This empirical validation rigorously bounds the DPU-visible covert channel to a noisy, sub-centibit-per-second trickle rather than a practical exfiltration path.

7.2 Fundamentally Invisible Attacks

Several attacks are invisible by construction to a metadata-only IDS, and no testbed configuration would change this:

- **A6 (training data poisoning):** occurs before gradients cross the wire; infrastructure-invisible by design.
- **A9 (memory bus snooping):** a passive electrical interposer leaves no software-visible metadata footprint.
- **A10 (passive timing observation):** an adversary who only watches emits no signal and creates no measurable deviation, the information-theoretic floor.
- **A13 (model/IP theft via side channels):** the in-scope passive variant behaves like A10; only an active probing variant (out of scope) would surface contention.
- **Norm-correct A5:** as established in Section 6.4, poisoning that preserves the gradient norm and timing leaves no metadata trace.

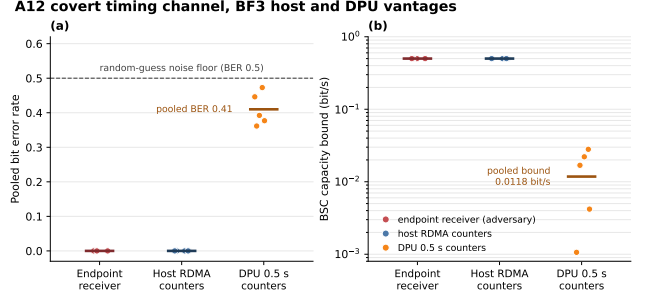


Fig. 9. A12 covert timing channel on the BF3 testbed; per-run points ($n = 5$ matched modulated repetitions) with a crossbar, by vantage. (a) Pooled bit error rate: the cooperating endpoint receiver and host RDMA counters decode the synthetic early/late burst channel perfectly (BER 0 every run), while the DPU 0.5 s counter telemetry sits just below the BER 0.5 random-guess noise floor (pooled BER 0.41, 266/649 slots). (b) Corresponding binary-symmetric-channel capacity bound: 0.5 bit/s at the endpoint and host collapses to 0.0118 bit/s at the DPU vantage (crossbar = bound from the pooled BER). An empirical demonstration of the detection noise floor, not a practical exfiltration path.

Future validation. A3 (routing manipulation, Tier 1) and A8 (TEE side channels, Tier 2) are predicted to be observable but remain unvalidated on our hardware: our topologies lack a multi-hop routing layer (A3), and neither testbed exposes a TEE to drive cache-probing leakage (A8), whose corroborating counters (H7-H9) sit inside the host boundary anyway. These reappear as future validation experiments in the agenda below.

Collectively, these limits scope the predictive surface precisely: a metadata-only IDS is a defense for integrity and availability threats (Tiers 1-2), not a universal solution. Payload content (A6), gradient direction (norm-correct A5), and passive observation (A10, A13) remain opaque, exactly as confidential computing intends.

8. OPEN PROBLEMS AND DEFENDER GUIDANCE

The unvalidated portions of the observability surface form the core research agenda for intrusion detection in confidential AI clusters. We organize them into open problems and the corresponding design guidance our current evidence already supports.

8.1 Open Problems

1. **Switch-backed congestion validation.** The congestion-cascade rows (N6, N8) are encryption-robust but topology-dependent; our back-to-back nulls leave their attack-time behavior on a real switched RDMA fabric unmeasured. Validating ECN/CNP under A1/A2 on a queuing fabric, across severities, is the highest-value next experiment.
2. **Multi-hop routing (A3).** Detecting routing manipulation via RTT anomaly and path asymmetry requires a multi-tenant, multi-hop testbed that neither of our vantages provided.
3. **TEE side channels (A8).** Staging cache-probing leakage against a real TDX/SEV enclave and corre-

lating it with in-TEE H7-H9 counters would close the most important Tier-2 gap.

4. **Cross-vantage fusion under a unified clock.** Our two vantages were measured independently; a fused testbed that exposes both the DPU plane and an in-TEE exporter on the same job would let attacks like A11 be correlated rather than stitched.
5. **Higher-severity and line-rate encrypted regimes.** A1 was measured only at low severity, and encryption invariance only at a host-software-ESP-limited 800 Mbit/s; confirming both at line rate and under DPU inline-IPsec offload remains open. DPU-accelerated detection and mitigation pipelines that already operate at these rates [18] offer a natural target for this confirmation.

8.2 Defender Design Guidance

Our worked examples already justify several concrete recommendations for building an IDS in this setting:

- **Place the primary network/infrastructure sensor on the DPU.** It is on the data path, architecturally outside the host TEE (so a compromised host cannot suppress it), and, per Section 6.2, its counters are encryption-invariant.
- **Key network detectors on the traffic envelope, not the congestion cascade.** Volume and flow-shape signals (N2-N5, H2, H5) are robust to both encryption and topology; treat ECN/CNP (N6, N8) as a bonus available only on queuing fabrics.
- **Exploit training symmetry at the workload layer.** Because data-parallel baselines are quiet and periodic, simple peer-spread and rank-spread statistics on W1/W5 are sufficient to flag A4/A7/A11; expose W6 if poisoning detection is in scope.
- **Require a late-fusion correlator across vantages.** Network attacks are DPU-dominated, training disruption is workload-dominated, and insider threats (A11) need both; a single vantage cannot distinguish a slow node from a fabric DoS.
- **Respect the cross-vantage discipline.** Do not extrapolate DPU evidence to GPU/NCCL effects or workload evidence to DPU flow state; these vantage lanes are also the detector’s integrity constraints.

9. RELATED WORK

Our systematization sits at the intersection of five bodies of work.

RDMA and network security. ReDMArk [3] documented fundamental vulnerabilities in RDMA deployments, and LoRDMA [4] introduced a low-rate DoS that exploits DCQCN [5] congestion control while evading rate-based detection; RoCEv2-at-scale experience [6] characterizes the PFC/ECN behavior these attacks ride on. We systematize the metadata footprint of such attacks (A1, A2) specifically under confidential transport, contributing the encryption-invariance evidence at the DPU vantage.

Distributed-training vulnerabilities. Byzantine-robust aggregation (Krum [7]) and the attacks that defeat it (the hidden vulnerability of distributed learning [8], “A Little Is Enough” [9], local model poisoning [10], and backdoor injection [11]) motivate our A4/A5 expectations. Unlike that line of work, we study not model convergence but the infrastructure- and workload-timing metadata these attacks emit, and we record where norm-correct poisoning escapes metadata entirely.

Cluster observability and telemetry. Holmes [12] and Mycroft [13] diagnose LLM-training clusters using NCCL tracing and GPU telemetry, while NetVigil [19] and Kitsune [20] perform metadata-only and unsupervised anomaly detection in data centers. These tools target reliability in trusted environments; we repurpose their sensitivity toward adversarial detection and toward the DPU as an isolated, encryption-robust vantage.

IDS over encrypted data and confidential computing. A complementary line attempts intrusion detection directly over encrypted traffic, including homomorphic-encryption and FHE-based detectors [21]–[24] and detection over encrypted network data [22]. These approaches pay heavy cryptographic cost to inspect content; our systematization instead shows how much can be recovered from metadata alone at an isolated vantage, sidestepping payload access. We also draw on counter-based detection [15] and on the defensive inversion of NVLink side channels [16], [17] as indicators.

DPU- and SmartNIC-accelerated IDS. A recent line of work builds intrusion detection and mitigation directly on data-processing units and SmartNICs: telemetry-driven DDoS detection in P4-based 5G user-plane functions on a DPU [25], DPUs with embedded GPUs performing local AI-based monitoring and security on the optical edge [26], hardware-accelerated intrusion prevention that mitigates line-rate attack traffic on a DPU [18], and SmartNIC-offloaded graph-neural-network IDS [27]. This work establishes the DPU/SmartNIC as a practical, line-rate detection host; our systematization is complementary, characterizing *what* such a detector can observe from the isolated DPU vantage once payloads are encrypted and the TEE boundary is in place.

10. CONCLUSION

The move to confidential computing secures the payloads of AI training clusters but appears, at first glance, to make them unmonitorable. This paper systematized why that appearance is misleading. Treating observability itself as the object of study, we established the predictive footprint each attack is expected to leave at every metadata layer, and selectively validated these predictions with empirical measurements. The network and infrastructure layers are predicted to retain a recoverable traffic envelope at the architecturally isolated DPU vantage, and we validated that this envelope is encryption-invariant within ESP framing overhead. The workload layer turns the symmetry

of distributed training into a sensitive, validated detector of disruption. We explicitly charted the theoretical limits, from covert-channel noise floors to fundamentally invisible passive, physical, and pre-wire attacks. The unvalidated predictions form a concrete research agenda, and the validated elements already provide a foundation for the design of cross-layer, DPU-vantage intrusion detection. Confidential AI clusters are not inherently unmonitorable; they are monitorable in specific, predictable ways, and this systematization is intended as the framework for the detectors that exploit them.

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